SeismicNet: A One-Dimensional Convolutional Neural Network for Automatic P- and S-Wave Arrival Picking in Local Seismic Records

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Abstract

Automatic phase picking is a central step in earthquake monitoring and seismic event characterization. We present SeismicNet, a one-dimensional convolutional neural network designed to detect P- and S-wave arrivals in local seismic waveform traces. The model receives a normalized 3000-sample trace and outputs three time-dependent probability distributions corresponding to P arrival, S arrival, and noise/background. Training labels are generated from manual picks by centering Gaussian distributions at the recorded arrival times. SeismicNet was trained using local seismic records from 2019–2020 and evaluated on unseen data from January and February 2021. Predictions were assessed by the temporal error between the maximum predicted probability and the manual phase label under 1 s, 2 s, and 3 s thresholds. The results show that the model learns meaningful phase-arrival patterns and performs better on less noisy data, while multi-event traces and swarm/volcanic activity remain major sources of error. These findings support the use of convolutional neural networks as assistance tools for routine seismic analysis and motivate larger, cleaner, and more diverse local waveform datasets for future automatic picking systems.

1 Introduction

Seismic phase picking is a fundamental task in routine earthquake analysis. Manual identification of P- and S-wave arrivals is accurate but time-consuming, motivating automatic methods that can support analysts and accelerate event characterization. This paper presents SeismicNet, a convolutional neural network developed for automatic phase detection in local seismic traces.

2 Background

2.1 Automatic seismic phase picking

2.2 Deep learning for seismic waveforms

2.3 Physics motivation

3 Data

3.1 Data source

The model was developed using local seismic records from INSIVUMEH. Training data correspond to 2019–2020, while January and February 2021 were used for evaluation on unseen data.

3.2 Preprocessing

Each selected waveform trace was resampled to a fixed dimension $d = 3000$ and normalized. Only traces with usable P and S picks inside the waveform interval were retained.

3.3 Label construction

Manual phase picks were transformed into sample indices and represented as Gaussian probability distributions centered on the P and S arrivals. A third channel represents noise/background. Therefore, for each input

$$x \in \mathbb{R}^{3000\times 1}, \tag{1}$$

the target is

$$Y \in \mathbb{R}^{3000\times 3}, \tag{2}$$

where the three output classes are P phase, S phase, and noise.

4 Methods

4.1 Architecture

SeismicNet uses one-dimensional convolution and transposed convolution layers followed by a dense projection and softmax output. The output is a probability distribution over the three classes for every temporal sample.

Table 1: SeismicNet architecture.

Layer	Parameters	Purpose
Conv1D	175 filters, kernel 7, ReLU	Local waveform features
MaxPooling1D	pool size 9	Temporal compression
Conv1D	125 filters, kernel 7, ReLU	Higher-level features
Conv1DTranspose	20 filters, kernel 7, stride 4	Temporal upsampling
Conv1DTranspose	25 filters, kernel 7, stride 2	Temporal upsampling
Conv1DTranspose	15 filters, kernel 8, stride 2	Temporal upsampling
MaxPooling1D	pool size 4	Output length control
Dense	9000 linear units	Projection to 3000×3
Reshape	$(3000, 3)$	Sequence output
Softmax	axis 2	P/S/noise probabilities

4.2 Training

The model was implemented in TensorFlow/Keras, trained with Adam, and optimized with categorical cross-entropy. Training was configured for up to 75 epochs with a batch size of 1500.

4.3 Inference

For each trace, the predicted P and S channels are searched for peaks. The highest P and S peaks define the predicted phase arrivals. If the predicted P index appears after the predicted S index, the order is swapped to enforce the expected physical ordering.

4.4 Evaluation

Prediction errors are defined as

$$\Delta t_P = \left| \frac{(i_P^{pred} - i_P^{label})}{d} (t_{end} - t_{start}) \right|, \tag{3}$$

$$\Delta t_S = \left| \frac{(i_S^{pred} - i_S^{label})}{d} (t_{end} - t_{start}) \right|, \tag{4}$$

where $d = 3000$ is the fixed trace dimension. We report correct picks under $\Delta t < 1$ s, $\Delta t < 2$ s, and $\Delta t < 3$ s.

5 Results

5.1 January 2021

5.2 February 2021

Table 2: SeismicNet results for February 2021.

Criterion	P phases	S phases	At least one phase	Both phases
$\Delta t < 1$ s	763	876	1172	467
$\Delta t < 2$ s	949	1161	1376	734
$\Delta t < 3$ s	1039	1277	1445	871

5.3 Qualitative prediction examples

6 Discussion

SeismicNet learned meaningful probability distributions for P- and S-wave arrivals. The model performed better on cleaner data and degraded on traces containing multiple possible events, swarm activity, or strong noise. These results suggest that dataset quality and event complexity are central limitations for local automatic phase picking.

7 Limitations and Future Work

Future work should include larger training datasets, stronger baselines, reproducible code packaging, uncertainty estimates, confidence calibration, and experiments with residual, recurrent, and transformer-based architectures.

8 Conclusion

SeismicNet demonstrates the feasibility of automatic P- and S-wave arrival picking using a one-dimensional convolutional neural network trained on local seismic data. The model can support routine seismic analysis, especially as part of a larger analyst-in-the-loop workflow.